

The JFreeReport Class Library

Version 0.8.1

An Introduction

Written by David Gilbert

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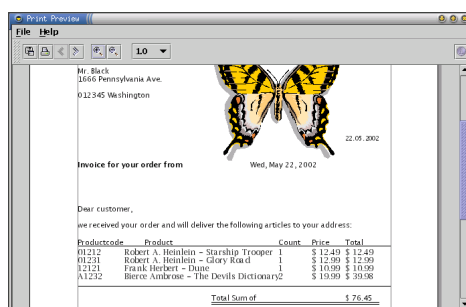
Contents

1	Introduction	2
1.1	What is JFreeReport?	2
1.2	The JFreeReport Project	2
1.3	This Document	2
1.4	Disclaimer	3
1.5	Acknowledgements	3
1.6	Comments and Suggestions	3
2	Downloading and Installing JFreeReport	4
2.1	Introduction	4
2.2	Download	4
2.3	Installation	4
2.4	Running the Demonstration Application	5
2.5	Compiling the Source	6
2.6	Generating the Javadoc Documentation	6
3	Getting Started	7
3.1	Introduction	7
3.2	The Basic Steps	7
3.3	Sample Application: Hello World	8
3.4	Sample Application: SwingIconsDemo	10
3.5	The Report Template File	12
3.6	Previewing the Report	29
4	Additional Documentation	31
4.1	Introduction	31
4.2	Reasons to Buy	31
A	The GNU Lesser General Public Licence	32
A.1	Introduction	32
A.2	The Licence	32

1 Introduction

1.1 What is JFreeReport?

JFreeReport is a free Java™ class library for generating reports. JFreeReport data is sourced via Swing's `TableModel` interface,¹ and formatted according to an XML-based report template file. Reports can be previewed on screen, sent to the printer or saved in various formats including PDF, HTML, Excel, CSV and plain text format.



JFreeReport is free under the terms of the GNU Lesser General Public Licence (LGPL)—see Appendix A for details.

1.2 The JFreeReport Project

The official home page of JFreeReport is:

<http://www.object-refinery.com/jfreereport/index.html>

Please visit this site for news, updates, and access to the JFreeReport forum.

Development of JFreeReport is now lead by Thomas Morgner. Development activities are coordinated through the project site hosted at SourceForge:

<http://sourceforge.net/projects/jfreereport/>

New developers are always welcome! If you would like to help out, please visit the project site and sign up.

1.3 This Document

This document has been written (by David Gilbert) for version 0.8.1 of JFreeReport.

Two versions of this document are available:

- the free version, which includes chapters up to and including the tutorial for new JFreeReport users.
- the premium version (available for purchase via the JFreeReport home page), which includes additional reference documentation for the XML report template file format and the JFreeReport classes.

¹Used to obtain data for tables created with Swing's `JTable` class.

The proceeds from the sale of the premium version of this document are used to sponsor on-going development of JFreeReport.

NOTE: the premium version of the documentation is temporarily unavailable while it is being updated for version 0.8.1.

1.4 Disclaimer

Please note that I cannot guarantee that this document is error free. You must use this document *at your own risk* or *not use it at all*.

1.5 Acknowledgements

I would like to thank Thomas Morgner for his (major) contribution as the JFreeReport Project Leader.

JFreeReport has also been enhanced by the contributions of several other developers: Piotr Bzdyl, Heiko Evermann, Patrice Rolland and Joerg Schoemer. The work by these contributors is very much appreciated.

Also, I would like to thank Bruno Lowagie and Paulo Soares for developing iText, the class library used in JFreeReport to save reports in PDF format.

Finally, thanks to all the developers that have provided feedback (bug reports and feature requests) concerning JFreeReport.

1.6 Comments and Suggestions

If you have any comments or suggestions regarding this document, please send an e-mail to: `david.gilbert@object-refinery.com`

2 Downloading and Installing JFreeReport

2.1 Introduction

This section contains instructions for:

- downloading and installing (unpacking) JFreeReport;
- running the demonstration applications;
- compiling² the JFreeReport source code;
- generating the Javadoc HTML documentation from the source code;

2.2 Download

You can download the latest version of JFreeReport from the JFreeReport project page on SourceForge:

<http://sourceforge.net/projects/jfreereport>

For version 0.8.1, you can choose between two download archives:

File:	Description:
<code>jfreereport-0.8.1.zip</code>	A <code>zip</code> file containing all the files required for JFreeReport (recommended for users of Microsoft Windows).
<code>jfreereport-0.8.1.tar.gz</code>	A <code>tar.gz</code> file containing all the files required for JFreeReport (recommended for users of Linux or Unix operating systems).

Both archives contain exactly the same source code, except that the end-of-line markers have been encoded using CR/LF in the `zip` archive, and LF only in the `tar.gz` archive.

Whichever archive you choose to download, you should save it in the directory where you want the `jfreereport-0.8.1` directory (created during unpacking—see the next section) to reside.

2.3 Installation

2.3.1 Overview

Installing JFreeReport is simply a matter of unpacking the download archive. When you unpack the archive, a new directory (`jfreereport-0.8.1`) will be created at the same location in your filesystem as the archive file itself.

2.3.2 Unpacking the ZIP Archive

If you chose to download the `jfreereport-0.8.1.zip` archive, you can “unzip” it using any ZIP utility. The `jar` utility included with Java will do the job—type the following command:

```
jar -xvf jfreereport-0.8.1.zip
```

²Recompilation is optional, because the precompiled runtime jar files are included in the download.

2.3.3 Unpacking the tar.gz Archive

If you chose to download the `jfreereport-0.8.1.tar.gz` archive, you can unpack it using the following command:

```
tar xvzf jfreereport-0.8.1.tar.gz
```

2.3.4 The Files

After unpacking the JFreeReport archive, you will see the following files and directories:

File:	Description:
<code>ChangeLog</code>	A text file containing notes on changes in each version of JFreeReport.
<code>README</code>	A text file containing important information about JFreeReport. Read this first!!!
<code>ant/</code>	A directory containing an Ant build script for JFreeReport.
<code>checkstyle/</code>	A directory containing a Checkstyle property file.
<code>jfreereport-0.8.1-demo.jar</code>	The runtime jar file for the JFreeReport demonstration applications.
<code>jfreereport-0.8.1.jar</code>	The runtime jar file for the JFreeReport class library.
<code>lib/</code>	A directory containing runtime jar files for the libraries that JFreeReport depends on.
<code>license-LGPL.txt</code>	The full text of the GNU Lesser General Public License.
<code>src/</code>	A directory containing the JFreeReport source code.

You should take a little time to familiarise yourself with the contents of the distribution and, in particular, read the `README` file.

2.4 Running the Demonstration Application

Several demonstration applications are included in the JFreeReport distribution. These are intended to illustrate some of the features supported by JFreeReport.

To run the main demo, switch to the `jfreereport-0.8.1` directory and type the following command:

```
java -jar jfreereport-0.8.1-demo.jar
```

This automatically sets up the appropriate classpath, and invokes the main demo class (`com.jrefinery.report.demo.JFreeReportDemo`). An alternative way to run the same application is to specify the classpath manually using the following command (all on one line):

```
java -classpath jfreereport-0.8.1-demo.jar:jfreereport-0.8.1.jar
:lib/jcommon-0.7.2.jar:lib/itext-0.96.jar:lib/gnujexp.jar
:lib/bsh-1.2b6.jar:lib/pixie.jar:lib/jakarta-poi-1.5.1-final-20020615.jar
com.jrefinery.report.demo.JFreeReportDemo
```

Special points to note:

- if you are using Windows, you should use a semi-colon rather than a colon as the path separator, and a backward slash rather than a forward slash within paths;
- depending on your system setup, you may need to specify the full path for the `java` executable.

2.5 Compiling the Source

An Ant build script is included in the JFreeReport distribution, in the `ant` directory. This script has been tested using Ant version 1.5.1. For more information about Ant, refer to:

<http://jakarta.apache.org/ant>

To recompile the JFreeReport source code, you can use the following command:

```
ant compile
```

This creates a temporary `build` directory within the `jfreereport-0.8.1` directory, compiles the JFreeReport classes, then creates a new runtime jar file.

Similarly, to recompile the JFreeReport demonstration applications, you can use the following command:

```
ant compile-demo
```

This creates a temporary `build` directory within the `jfreereport-0.8.1` directory, compiles the JFreeReport demo classes, then creates a new runtime jar file.

2.6 Generating the Javadoc Documentation

The JFreeReport source code contains *Javadoc comments*. You can use the `javadoc` tool to generate HTML documentation files directly from the source code.³

To generate the documentation, you can use the Ant build script:

```
ant javadoc
```

This will create a new `javadoc` directory (located within the `jfreereport-0.8.1` directory) containing the Javadoc reference information for JFreeReport.

To view the Javadocs, open the `javadoc/index.html` file in your favourite web browser.

³The Javadoc HTML files can be viewed on-line at the JFreeReport home page.

3 Getting Started

3.1 Introduction

In this section, two sample applications are presented as an introduction to using JFreeReport:

- `HelloWorld.java` – a minimal application that illustrates just the basic steps in creating and displaying a report;
- `SwingIconsDemo.java` – a more complete example that demonstrates the use of various formatting options provided by JFreeReport;

These examples should be sufficient to get you started using JFreeReport in your own applications.

3.2 The Basic Steps

3.2.1 Overview

In standard usage, there are three major tasks in generating reports with JFreeReport:

- arrange for some data that can be accessed via the `TableModel` interface (that is, the model used by Swing's `JTable` class);
- create a `JFreeReport` object that will control the formatting of the generated report;
- link the data (`TableModel`) with the `JFreeReport` instance and pass the report to a print preview frame for presentation to the user;

Other variations are possible (for example, sending a report directly to file *without* previewing it first) but the above steps represent the most common scenario.

3.2.2 Step 1: The Data

JFreeReport is designed to work with data that is accessible via the `TableModel` interface. If you are already familiar with using Swing's `JTable` class, then this will cause you no trouble. If not, I recommend that you find out more about Swing's `JTable` class before you tackle JFreeReport. Tutorials can be found at Sun's Java website:

<http://java.sun.com/>

If you want to generate reports using data accessed via JDBC, you will be pleased to know that JFreeReport includes code for generating a `TableModel` instance from a JDBC `ResultSet`.

3.2.3 Step 2: The Report Definition

All report formatting information is recorded by an instance of the `JFreeReport` class. You have two options for creating this instance:

- write a report definition in XML format, and get `JFreeReport` to parse the definition and create a corresponding `JFreeReport` object;
- create a `JFreeReport` instance in code, and populate the report bands and elements using the `JFreeReport` API;

Putting your report definitions into an XML file is a good idea, because it allows you to change the formatting of your reports without recompiling your application.

Creating reports in code can allow additional flexibility if you want to vary the format of your reports at run-time, but has the disadvantage that your report formatting is “hard-coded” into your application.

3.2.4 Step 3: Previewing the Report

The final step—previewing the report—is mostly taken care of by the `JFreeReport` library. The print preview frame will display the report on-screen, provide controls to page forward and backward through the report, zoom in and zoom out, print the report, or export to a number of different file formats including PDF, HTML, CSV, Excel and XML.

3.3 Sample Application: Hello World

3.3.1 Overview

The `HelloWorld.java` application (included in the distribution) provides a basic overview of the steps for creating and viewing reports with `JFreeReport`.

3.3.2 The Data

The data for this example is a simple table:

Column1:	Column2:
Hello	World!

It is created using the following code:

```
private TableModel createData()
{
    Object[] columnNames = new String[] { "Column1", "Column2" };
    DefaultTableModel result = new DefaultTableModel(columnNames, 1);
    result.setValueAt("Hello", 0, 0);
    result.setValueAt("World!", 0, 1);
    return result;
}
```

The column names are important, since they will be referenced by elements within the report definition (see the next section).

3.3.3 The Report

The `JFreeReport` instance used to control the format of the “Hello World” report is created in code, so that the demo is self-contained:

```
private JFreeReport createReportDefinition()
{
    JFreeReport report = new JFreeReport();
    report.setName("A Very Simple Report");

    TextElement t1 = ItemFactory.createStringElement(
        "T1",
        new Rectangle2D.Double(0.0, 0.0, 150.0, 20.0),
        Color.black,
        ElementAlignment.LEFT.getOldAlignment(),
        ElementAlignment.MIDDLE.getOldAlignment(),
        null, // font
        "-", // null string
        "Column1"
    );

    report.getItemBand().addElement(t1);

    TextElement t2 = ItemFactory.createStringElement(
        "T2",
        new Rectangle2D.Double(200.0, 0.0, 150.0, 20.0),
        Color.black,
        ElementAlignment.LEFT.getOldAlignment(),
        ElementAlignment.MIDDLE.getOldAlignment(),
        null, // font
        "-", // null string
        "Column2"
    );

    report.getItemBand().addElement(t2);
    return report;
}
```

To understand what is going on here, you need to know that the report layout is controlled by *elements* that are added to *bands* within the report. A report consists of a number of (possibly empty) bands, including:

- the *report header* – printed once at the beginning of the report;
- the *report footer* – printed once at the end of the report;
- the *page header* – printed at the top of each page;
- the *page footer* – printed at the bottom of each page;
- the *item band* – printed once for each row of data in the `TableModel`;
- an additional *group header* and/or *group footer* for each *group* defined in the report.

In the `HelloWorld.java` application, only the item band is populated—the other bands remain empty (the default). A text element is added for each column of data in the `TableModel`.

The `ItemFactory` class is used to create individual elements. Each element carries its own position, alignment, font settings and other presentation attributes. Notice also that each element in the example is tied back to the `TableModel` by a reference to a column name in the table.

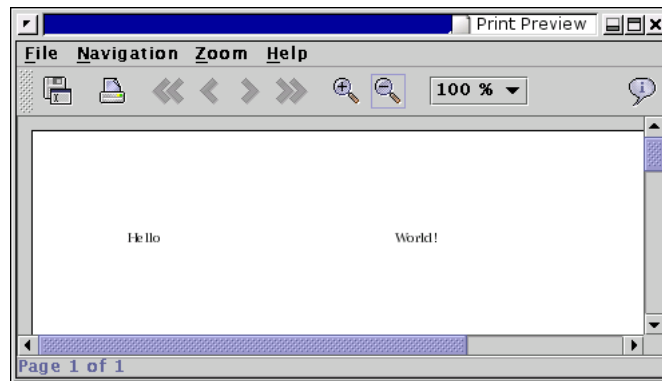
3.3.4 Previewing the Report

To preview this report, the data is linked with the report (using the `setData(...)` method), and the report is passed to a print preview frame:

```
TableModel data = createData();
JFreeReport report = createReportDefinition();

report.setData(data);
try
{
    PreviewFrame preview = new PreviewFrame(report);
    preview.pack();
    preview.setVisible(true);
}
catch (ReportProcessingException e)
{
    Log.error("Failed to generate report ", e);
}
```

If you run the demo, you should see the following preview frame:



Note: to run the demo from the precompiled jar files in the JFreeReport distribution, you can use the following command:

```
java -classpath jfreereport-0.8.1-demo.jar
:jfreereport-0.8.1.jar:lib/jcommon-0.7.2.jar
:lib/gnujarp.jar:lib/bsh-1.2b6.jar:lib/pixie.jar
:lib/itext-0.96.jar:lib/jakarta-poi-1.5.1-final-20020615.jar
com.jrefinery.report.demo.HelloWorld
```

As usual, colons should be replaced by semi-colons if you are using Windows.

3.4 Sample Application: SwingIconsDemo

3.4.1 Overview

The `SwingIconsDemo.java` application (included in the distribution) generates a report listing the icons included in Sun's *Java Look and Feel Graphics Repository*, a collection of standard icons distributed by Sun Microsystems for use with Java. You will need to download this icon set, and agree to Sun's terms and conditions, before you will be able to run the demo application.

This example uses an XML report definition file to create the `JFreeReport` instance that controls the final report output.

3.4.2 Running the Demo

You may find the material in the following sections easier to follow if you have already tried out the demo application. In this section I describe how to run the `SwingIconsDemo` application.

The `SwingIconsDemo.java` source file is included in the JFreeReport distribution, along with the report template file (`swing-icons.xml`).

As mentioned previously, running this demo requires you to first download the *Java Look and Feel Graphics Repository*, a collection of standard icons (provided by Sun) for use in Swing applications. These icons are contained in a jar file (`jlfgr-1.0.jar`) that you can download⁴ from:

<http://developer.java.sun.com/developer/techDocs/hi/repository/>

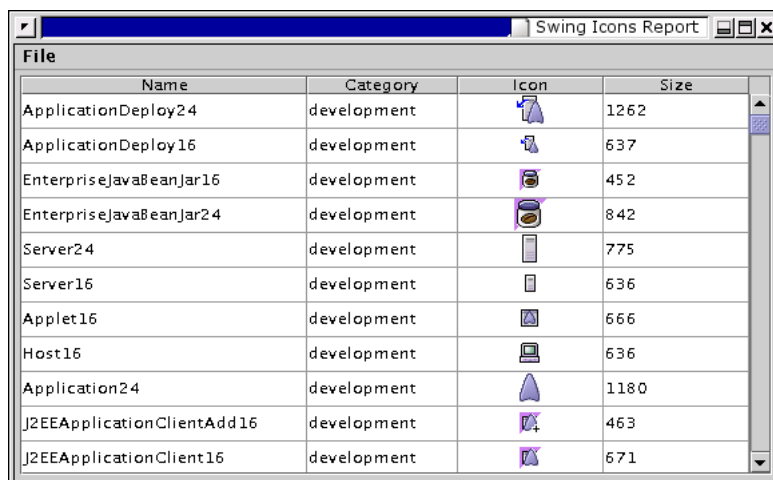
When you run the `SwingIconsDemo` application, it will look for this file in a *directory* that is on the class path. I recommend that you place the file in the `jfreereport-0.8.1` directory, then use the following command to run the demo:

```
java -classpath .:jfreereport-0.8.1-demo.jar
:jfreereport-0.8.1.jar:lib/jcommon-0.7.2.jar
:lib/gnujaxp.jar:lib/bsh-1.2b6.jar:lib/pixie.jar
:lib/iText-0.96.jar:lib/jakarta-poi-1.5.1-final-20020615.jar
com.jrefinery.report.demo.SwingIconsDemo
```

Notice the addition of the “.” at the beginning of the classpath to include the current directory in the classpath.

3.4.3 The Data

The `SwingIconsDemo` class uses the `SwingIconsDemoTableModel` class to store the data for the report. This table model contains code for reading the icons from the icon file (`jlfgr-1.0.jar`). The data is then displayed in a simple user interface using a `JTable`:



The screenshot shows a window titled "Swing Icons Report". Inside the window is a table with four columns: "Name", "Category", "Icon", and "Size". The table lists various Java Swing icons, all categorized as "development".

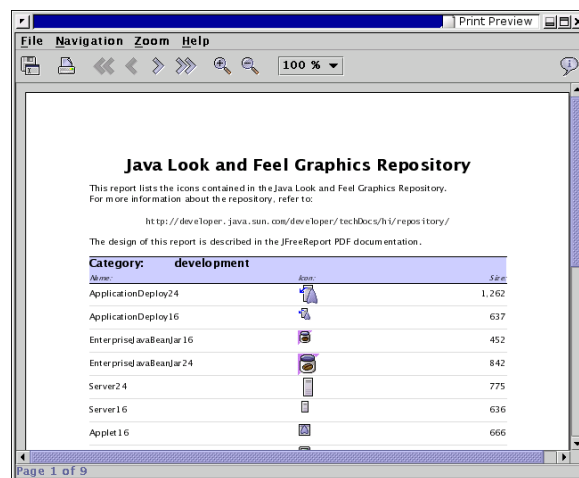
Name	Category	Icon	Size
ApplicationDeploy24	development		1262
ApplicationDeploy16	development		637
EnterpriseJavaBeanJar16	development		452
EnterpriseJavaBeanJar24	development		842
Server24	development		775
Server16	development		636
Applet16	development		666
Host16	development		636
Application24	development		1180
J2EEApplicationClientAdd16	development		463
J2EEApplicationClient16	development		671

⁴Subject to acceptance of Sun's licence agreement.

Some important points to note about the dataset:

- the column names are *Name*, *Category*, *Icon* and *Size*. These names are used in the XML report template file to reference data items;
- the data is sorted by the *Category* column. This is important since the data will be grouped by *Category*. JFreeReport does not currently perform any sorting, so you need to provide the data pre-sorted.

When you run the `SwingIconsDemo` application, you will see a menu item that allows you to display a print preview window. When you preview the sample report, you will see output like this:



The format of the report is controlled by an XML report template file (`swing-icons.xml`), described in the next section.

3.5 The Report Template File

3.5.1 Overview

The `swing-icons.xml` file contains a report definition in the format required by JFreeReport's "extended" parser (a new feature in version 0.8.1). This parser is more powerful and flexible than the former "simple" parser, at the expense of being more verbose. The old format is still supported for backward compatibility, and a utility is included with JFreeReport to convert reports from the old format to the new format.

Here is a skeleton version of the file (the missing pieces, marked with comments, are reproduced in later sections):

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE report PUBLIC "-//JFreeReport/DTD report definition//EN//extended"
    "http://jfreereport.sourceforge.net/extreport.dtd">

<report-definition name="First Report">

    <!-- * INSERT PARSER CONFIGURATION HERE * -->

    <!-- * INSERT REPORT CONFIGURATION HERE * -->

    <!-- * INSERT STYLES HERE * -->

    <!-- * INSERT TEMPLATES HERE * -->
```

```

<!-- * INSERT FUNCTIONS HERE * -->

<!-- ***** -->
<!-- * REPORT DESCRIPTION * -->
<!-- ***** -->

<report-description>

    <!-- INSERT REPORT HEADER HERE -->

    <!-- INSERT REPORT FOOTER HERE -->

    <!-- INSERT PAGE HEADER HERE -->

    <!-- INSERT PAGE FOOTER HERE -->

    <!-- INSERT GROUP DEFINITIONS HERE -->

    <!-- INSERT ITEM BAND HERE -->

</report-description>

</report-definition>

```

The report name is required, and is made available within the report as a *report property* (`report.name`).

3.5.2 The Parser Configuration

The *parser configuration* provides a mechanism for customising the behaviour of the report definition parser:

```

<!-- ***** -->
<!-- * PARSER CONFIGURATION * -->
<!-- ***** -->
<parser-config>
  <object-factory class="com.jrefinery.report.io.ext.factory.datasource.DataSourceCollector"/>
  <object-factory class="com.jrefinery.report.io.ext.factory.objects.DefaultClassFactory"/>
  <element-factory class="com.jrefinery.report.io.ext.factory.elements.DefaultElementFactory"/>
  <stylekey-factory class="com.jrefinery.report.io.ext.factory.stylekey.DefaultStyleKeyFactory"/>
  <stylekey-factory class="com.jrefinery.report.io.ext.factory.stylekey.PageableLayoutStyleKeyFactory"/>
  <template-factory class="com.jrefinery.report.io.ext.factory.templates.DefaultTemplateCollection"/>
  <datasource-factory class="com.jrefinery.report.io.ext.factory.datasource.DefaultDataSourceFactory"/>
</parser-config>

```

Most of the time, you will just copy and paste this default configuration.

3.5.3 The Report Configuration

The *report configuration* allows you to control certain aspects of the way that JFreeReport works. In the example, the default page format and the initial height and width of the preview frame is specified:

```

<!-- ***** -->
<!-- * REPORT CONFIGURATION * -->
<!-- ***** -->
<report-config>
  <defaultpageformat orientation="portrait"
    pageformat="LETTER"
    topmargin="72"
    bottommargin="72"
    leftmargin="72"
    rightmargin="72"/>

  <configuration>
    <property name="com.jrefinery.report.preview.PreferredHeight">480</property>
    <property name="com.jrefinery.report.preview.PreferredWidth">640</property>
  </configuration>
</report-config>

```

All the valid `pageformat` options are specified in the `PageFormatFactory` class. The margins are specified in points (1/72 inch).

Additional details on the available configuration parameters can be found in the documentation for the [ReportConfiguration](#) class.

3.5.4 Styles

The *Styles* element is empty in this example:

```
<!-- ***** -->
<!-- * STYLES * -->
<!-- ***** -->
<styles>
</styles>
```

3.5.5 Templates

The *Templates* element is empty in this example:

```
<!-- ***** -->
<!-- * TEMPLATES * -->
<!-- ***** -->
<templates>
</templates>
```

3.5.6 Report Functions

The example uses several *report functions* to calculate summary information for display on the report. It is necessary to declare functions within the XML report template file—individual report elements can then reference particular functions by name.

When you declare a function, you give it a name and specify the Java class that implements the function. Most functions also require some properties to be specified (you should refer to the documentation or source code for the class that implements the function to find out which properties are required).

Here is the XML used to define the functions used in the sample report:

```
<!-- ***** -->
<!-- * FUNCTIONS * -->
<!-- ***** -->
<functions>

  <property-ref name="report.date"/>

  <function name="PageNumber" class="com.jrefinery.report.function.PageFunction">
  </function>

  <function name="GroupCount" class="com.jrefinery.report.function.ItemCountFunction">
    <properties>
      <property name="field">Name</property>
      <property name="group">Category</property>
    </properties>
  </function>

  <function name="GroupSum" class="com.jrefinery.report.function.ItemSumFunction">
    <properties>
      <property name="field">Size</property>
      <property name="group">Category</property>
    </properties>
  </function>

  <function name="OverallCount" class="com.jrefinery.report.function.ItemCountFunction">
    <properties>
      <property name="field">Name</property>
    </properties>
  </function>

  <function name="OverallSum" class="com.jrefinery.report.function.ItemSumFunction">
    <properties>
      <property name="field">Size</property>
    </properties>
  </function>
</functions>
```

The first entry in the function declarations marks a report property for access. The (**report.date**) property is automatically set up by JFreeReport at report generation time, and it returns the current date. In the example, the report date is displayed in the page header.

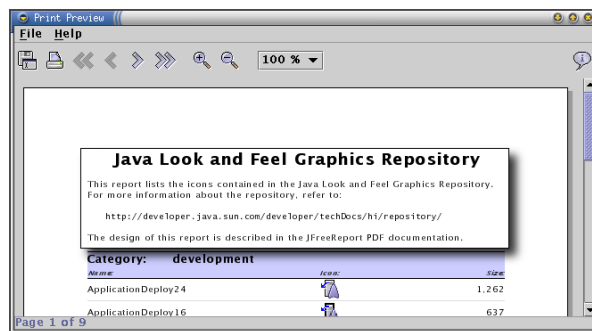
The first function (`PageNumber`) returns the current page number. It does not require any properties to be set. In the example report, the page number is displayed in the page footer.

There are two functions declared to count report items. The first (`GroupCount`) counts items within the `Category` group, and is used in the group footer. The second (`OverallCount`) counts items for the entire report (simply by omitting the `group` property), and is used in the report footer.

There are two functions declared to sum the icon size field. The first (`GroupSum`) calculates a running total within the `Category` group, and is used in the group footer. The second (`OverallSum`) calculates a running total for the entire report, and is used in the report footer.

3.5.7 The Report Header

The *report header* is a band that is printed once at the beginning of a report. In our example, the report header contains five labels (fixed text items), highlighted in the figure below:



The minimum height of the band is 112 points. The default font is plain `SansSerif`, at 10 points. Here is the XML used to define the report header:

```
<!-- ===== -->
<!-- = REPORT HEADER = -->
<!-- ===== -->
<report-header name="report-header-band">

  <style>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">112.0</basic-object>
      <basic-object name="width" class="java.lang.Double">0.0</basic-object>
    </compound-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-size">10</basic-key>
    <basic-key name="font-bold">>false</basic-key>
    <basic-key name="font-italic">>false</basic-key>
    <basic-key name="pagebreak-after">>false</basic-key>
  </style>

  <default-style>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-size">10</basic-key>
    <basic-key name="font-bold">>false</basic-key>
    <basic-key name="font-italic">>false</basic-key>
  </default-style>

  <element name="title1" type="text/plain">
    <style>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">2.0</basic-object>
      </compound-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">18.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
      </compound-key>
      <basic-key name="dynamic_height">>false</basic-key>
    </style>
  </element>
</report-header>
```



```

<basic-key name="alignment">center</basic-key>
<basic-key name="valignment">bottom</basic-key>
<basic-key name="font-size">18</basic-key>
<basic-key name="font-bold">true</basic-key>
<basic-key name="font-italic">false</basic-key>
<basic-key name="paint">black</basic-key>
</style>
<template references="label">
  <basic-object name="content">Java Look and Feel Graphics Repository</basic-object>
  <basic-object name="nullValue">null</basic-object>
</template>
</element>

<element name="description1" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">32.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">10.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="dynamic_height">false</basic-key>
    <basic-key name="alignment">left</basic-key>
    <basic-key name="valignment">bottom</basic-key>
    <basic-key name="paint">black</basic-key>
  </style>
  <template references="label">
    <basic-object name="content">This report lists the icons contained in the Java Look and Feel Graphics Repository.</basic-object>
    <basic-object name="nullValue">null</basic-object>
  </template>
</element>

<element name="description2" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">44.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">10.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="dynamic_height">false</basic-key>
    <basic-key name="alignment">left</basic-key>
    <basic-key name="valignment">bottom</basic-key>
    <basic-key name="paint">black</basic-key>
  </style>
  <template references="label">
    <basic-object name="content">For more information about the repository, refer to:</basic-object>
    <basic-object name="nullValue">null</basic-object>
  </template>
</element>

<element name="url1" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">68.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">10.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="dynamic_height">false</basic-key>
    <basic-key name="alignment">center</basic-key>
    <basic-key name="valignment">bottom</basic-key>
    <basic-key name="font">Monospaced</basic-key>
    <basic-key name="font-size">9</basic-key>
    <basic-key name="paint">black</basic-key>
  </style>
  <template references="label">
    <basic-object name="content">http://developer.java.sun.com/developer/techDocs/hi/repository/</basic-object>
    <basic-object name="nullValue">null</basic-object>
  </template>
</element>

<element name="description3" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">92.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">10.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="dynamic_height">false</basic-key>
    <basic-key name="alignment">left</basic-key>
    <basic-key name="valignment">bottom</basic-key>
    <basic-key name="paint">black</basic-key>
  </style>
  <template references="label">
    <basic-object name="content">The design of this report is described in the JFreeReport PDF documentation.</basic-object>
    <basic-object name="nullValue">null</basic-object>
  </template>
</element>

```

</report-header>

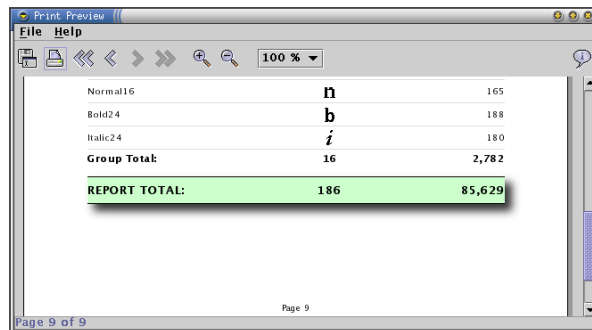
Special points to note:

- the x and y coordinates for each element are measured in points (1/72 inch) relative to the band's origin;
- the y-values increase as you move down the band / page;
- the width of each label is 100 percent (percentages are coded as negative values), which means that the element area extends across the full width of the report. This is particularly important for the first label, which is centered within its area, and so appears at the center of the band;
- the default font is overridden for first and fourth labels. For the other labels, the default font specified for the header is used.

Although the example displays only labels, you can use any report element in a report header, including those that display function values.

3.5.8 The Report Footer

The *report footer* is a band that is printed once at the end of a report. In the example, the report footer contains a label and two number fields that get their values from specific report functions. It also contains two lines and a rectangle, use to enhance the appearance of the report. The footer is highlighted in the figure below:



Here is the XML used to define the report footer:

```
<!-- ===== -->
<!-- = REPORT FOOTER = -->
<!-- ===== -->
<report-footer name="report-footer-band">

  <style>
    <basic-key name="font-bold">true</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">30.0</basic-object>
      <basic-object name="width" class="java.lang.Double">0.0</basic-object>
    </compound-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-size">12</basic-key>
    <basic-key name="font-italic">>false</basic-key>
    <basic-key name="pagebreak-before">>false</basic-key>
  </style>

  <default-style>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-size">12</basic-key>
    <basic-key name="font-bold">true</basic-key>
    <basic-key name="font-italic">>false</basic-key>
  </default-style>
</report-footer>
```

```

</default-style>

<element name="rect1" type="shape/generic">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">0.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">-100.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="stroke" class="java.awt.BasicStroke">0.0</basic-key>
    <basic-key name="paint">#ccffcc</basic-key>
    <basic-key name="draw-shape">false</basic-key>
    <basic-key name="fill-shape">true</basic-key>
    <basic-key name="scale">true</basic-key>
    <basic-key name="keepAspectRatio">false</basic-key>
  </style>
  <datasource type="StaticDataSource">
    <compound-object name="value" class="java.awt.geom.Rectangle2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="height">100.0</basic-object>
      <basic-object name="width">100.0</basic-object>
      <basic-object name="y">0.0</basic-object>
    </compound-object>
  </datasource>
</element>

<element name="line1" type="shape/generic">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">0.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">0.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="paint">black</basic-key>
    <basic-key name="stroke" class="java.awt.BasicStroke">0.5</basic-key>
    <basic-key name="draw-shape">true</basic-key>
    <basic-key name="fill-shape">false</basic-key>
    <basic-key name="keepAspectRatio">false</basic-key>
    <basic-key name="scale">true</basic-key>
  </style>
  <datasource type="StaticDataSource">
    <compound-object name="value" class="java.awt.geom.Line2D$Float">
      <basic-object name="x2">100.0</basic-object>
      <basic-object name="x1">0.0</basic-object>
      <basic-object name="y2">0.0</basic-object>
      <basic-object name="y1">0.0</basic-object>
    </compound-object>
  </datasource>
</element>

<element name="line2" type="shape/generic">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">30.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">0.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="stroke" class="java.awt.BasicStroke">0.5</basic-key>
    <basic-key name="paint">black</basic-key>
    <basic-key name="draw-shape">true</basic-key>
    <basic-key name="fill-shape">false</basic-key>
    <basic-key name="scale">true</basic-key>
    <basic-key name="keepAspectRatio">false</basic-key>
  </style>
  <datasource type="StaticDataSource">
    <compound-object name="value" class="java.awt.geom.Line2D$Float">
      <basic-object name="x2">100.0</basic-object>
      <basic-object name="x1">0.0</basic-object>
      <basic-object name="y2">30.0</basic-object>
      <basic-object name="y1">30.0</basic-object>
    </compound-object>
  </datasource>
</element>

<element name="reportTotalLabel" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">10.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">12.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-50.0</basic-object>
    </compound-key>
    <basic-key name="alignment">left</basic-key>
    <basic-key name="valignment">bottom</basic-key>
    <basic-key name="dynamic_height">false</basic-key>
    <basic-key name="paint">black</basic-key>
  </style>
  <template references="label">

```

```

        <basic-object name="content">REPORT TOTAL:</basic-object>
        <basic-object name="nullValue">null</basic-object>
      </template>
    </element>

    <element name="reportCountField" type="text/plain">
      <style>
        <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
          <basic-object name="x">-50.0</basic-object>
          <basic-object name="y">10.0</basic-object>
        </compound-key>
        <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
          <basic-object name="height" class="java.lang.Double">9.0</basic-object>
          <basic-object name="width" class="java.lang.Double">-30.0</basic-object>
        </compound-key>
        <basic-key name="paint">black</basic-key>
        <basic-key name="alignment">center</basic-key>
        <basic-key name="valignment">bottom</basic-key>
        <basic-key name="dynamic_height">>false</basic-key>
      </style>
      <template references="number-field">
        <basic-object name="field">OverallCount</basic-object>
        <basic-object name="format">#0</basic-object>
        <basic-object name="nullValue">-</basic-object>
      </template>
    </element>

    <element name="reportSumField" type="text/plain">
      <style>
        <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
          <basic-object name="x">-80.0</basic-object>
          <basic-object name="y">10.0</basic-object>
        </compound-key>
        <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
          <basic-object name="height" class="java.lang.Double">9.0</basic-object>
          <basic-object name="width" class="java.lang.Double">-20.0</basic-object>
        </compound-key>
        <basic-key name="paint">black</basic-key>
        <basic-key name="alignment">right</basic-key>
        <basic-key name="valignment">bottom</basic-key>
        <basic-key name="dynamic_height">>false</basic-key>
      </style>
      <template references="number-field">
        <basic-object name="field">OverallSum</basic-object>
        <basic-object name="format">#,##0</basic-object>
        <basic-object name="nullValue">-</basic-object>
      </template>
    </element>
  </report-footer>

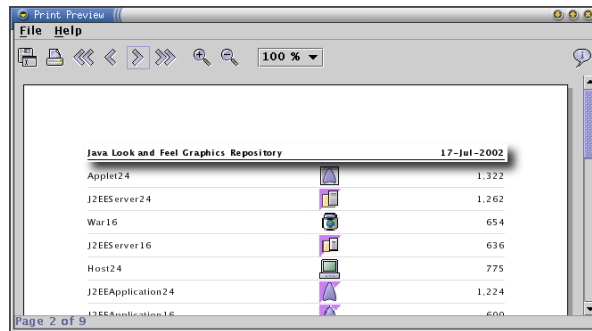
```

Special points to note:

- the lines and shapes are declared first, so that they are drawn first (to avoid obscuring the other report elements in the band);
- the coordinates for the **line** elements both represent a single point, in this example. This is a special case, where JFreeReport draws a horizontal line passing through the point and covering the full width of the band;
- the **number-field** elements specify the name of the function that supplies the element value. Each function must be declared elsewhere in the report definition (see section 3.5.6);
- for the **number-field** report elements, you can define a format string to format the output. Internally, this is passed to an instance of Java's `NumberFormat` class to control the formatting.

3.5.9 The Page Header

The *page header* appears at the top of every page, with the possible exception of the first and last pages. In the example, the page header contains a label and a date field that displays the current date. This is highlighted in the figure below:



Here is the XML used to define the page header:

```
<!-- ===== -->
<!-- = PAGE HEADER = -->
<!-- ===== -->
<page-header name="anonymousBand@1319318">

<style>
  <basic-key name="font-bold">true</basic-key>
  <basic-key name="display-on-lastpage">true</basic-key>
  <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
    <basic-object name="height" class="java.lang.Double">18.0</basic-object>
    <basic-object name="width" class="java.lang.Double">0.0</basic-object>
  </compound-key>
  <basic-key name="font-size">9</basic-key>
  <basic-key name="display-on-firstpage">>false</basic-key>
  <basic-key name="font">SansSerif</basic-key>
  <basic-key name="font-italic">>false</basic-key>
</style>

<default-style>
  <basic-key name="font-bold">true</basic-key>
  <basic-key name="font-size">9</basic-key>
  <basic-key name="font">SansSerif</basic-key>
  <basic-key name="font-italic">>false</basic-key>
</default-style>

<element name="@anonymous" type="shape/generic">
  <style>
    <basic-key name="stroke" class="java.awt.BasicStroke">0.0</basic-key>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">0.0</basic-object>
    </compound-key>
    <basic-key name="draw-shape">>false</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">-100.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="keepAspectRatio">>false</basic-key>
    <basic-key name="scale">true</basic-key>
    <basic-key name="paint">#afafaf</basic-key>
    <basic-key name="fill-shape">true</basic-key>
  </style>
  <datasource type="StaticDataSource">
    <compound-object name="value" class="java.awt.geom.Rectangle2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="height">100.0</basic-object>
      <basic-object name="width">100.0</basic-object>
      <basic-object name="y">0.0</basic-object>
    </compound-object>
  </datasource>
</element>

<element name="@anonymous" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">0.0</basic-object>
    </compound-key>
    <basic-key name="dynamic_height">>false</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">14.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-50.0</basic-object>
    </compound-key>
    <basic-key name="paint">black</basic-key>
    <basic-key name="alignment">left</basic-key>
    <basic-key name="valignment">bottom</basic-key>
  </style>
  <template references="label">
    <basic-object name="nullValue">>null</basic-object>
    <basic-object name="content">Java Look and Feel Graphics Repository</basic-object>
  </template>
</element>
```

```

<element name="@anonymous" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">-80.0</basic-object>
      <basic-object name="y">0.0</basic-object>
    </compound-key>
    <basic-key name="dynamic_height">false</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">14.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-20.0</basic-object>
    </compound-key>
    <basic-key name="paint">black</basic-key>
    <basic-key name="alignment">right</basic-key>
    <basic-key name="valignment">bottom</basic-key>
  </style>
  <template references="date-field">
    <basic-object name="nullValue">-</basic-object>
    <basic-object name="format">d-MMM-yyyy</basic-object>
    <basic-object name="field">report.date</basic-object>
  </template>
</element>

<element name="@anonymousf" type="shape/generic">
  <style>
    <basic-key name="stroke" class="java.awt.BasicStroke">2.0</basic-key>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">0.0</basic-object>
      <basic-object name="y">16.0</basic-object>
    </compound-key>
    <basic-key name="draw-shape">true</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">0.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
    </compound-key>
    <basic-key name="keepAspectRatio">false</basic-key>
    <basic-key name="scale">true</basic-key>
    <basic-key name="paint">#cfcfcf</basic-key>
    <basic-key name="fill-shape">false</basic-key>
  </style>
  <datasource type="StaticDataSource">
    <compound-object name="value" class="java.awt.geom.Line2D$Float">
      <basic-object name="x2">100.0</basic-object>
      <basic-object name="x1">0.0</basic-object>
      <basic-object name="y2">16.0</basic-object>
      <basic-object name="y1">16.0</basic-object>
    </compound-object>
  </datasource>
</element>
</page-header>

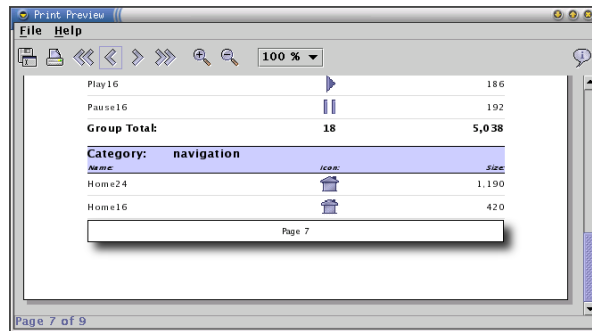
```

Special points to note:

- the page header is suppressed on the first page, since it would conflict with the report header;
- the `date-field` element references (by name) the report date property. This property is declared elsewhere in the report definition (see section 3.5.6);
- the `date-field` element has a format attribute to control date formatting. This is passed to an instance of Java's `SimpleDateFormat` class to control the formatting.

3.5.10 The Page Footer

The *page footer* appears at the bottom of every page, with the possible exception of the first and last pages. In the example, the page footer contains just a number field that displays the page number. This is highlighted in the figure below:



Here is the XML used to define the page footer:

```
<!-- ===== -->
<!-- = PAGE FOOTER = -->
<!-- ===== -->
<page-footer name="anonymousBand@7205769">

  <style>
    <basic-key name="font-bold">false</basic-key>
    <basic-key name="display-on-lastpage">true</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">14.0</basic-object>
      <basic-object name="width" class="java.lang.Double">0.0</basic-object>
    </compound-key>
    <basic-key name="font-size">9</basic-key>
    <basic-key name="display-on-firstpage">true</basic-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-italic">false</basic-key>
  </style>

  <default-style>
    <basic-key name="font-bold">false</basic-key>
    <basic-key name="font-size">9</basic-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-italic">false</basic-key>
  </default-style>

  <element name="@anonymous10" type="text/plain">
    <style>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">3.0</basic-object>
      </compound-key>
      <basic-key name="dynamic_height">false</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">9.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
      </compound-key>
      <basic-key name="paint">black</basic-key>
      <basic-key name="alignment">center</basic-key>
      <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="number-field">
      <basic-object name="nullValue">null</basic-object>
      <basic-object name="format">Page #0</basic-object>
      <basic-object name="field">PageNumber</basic-object>
    </template>
  </element>
</page-footer>
```

Special points to note:

- the page footer can be suppressed on the first or last pages, to avoid conflict with the report header and/or the report footer. In this example, the page footer is displayed on all pages;
- the elements using **number-field** templates reference (by name) the function that returns the page number. This function is declared elsewhere in the report definition (see section 3.5.6);
- the **number-field** element has a **format** attribute to control numberformatting. This is passed to an instance of Java's **NumberFormat** class to control formatting.

3.5.11 Report Groups

Report groups are used to aggregate report items according to the values in certain fields. In this example, only one group is defined (on the **Category** field) in addition to the default group, but it is possible to create multiple (nested) groups with JFreeReport.

For each group, you can define a *group header* that is printed at the start of each group instance. Likewise, you can define a *group footer* that is printed at the end of each group instance.

The XML used to define the **Category** group is:

```
<!-- ===== -->
<!-- = GROUPS = -->
<!-- ===== -->
<groups>

  <!-- default group -->
  <group name="default">
    <fields>
    </fields>
    <group-header name="anonymousBand@4732779">
      <style>
      </style>
    </group-header>
    <group-footer name="anonymousBand@6100951">
      <style>
      </style>
    </group-footer>
  </group>

  <!-- category group -->
  <group name="Category">
    <fields>
    <field>Category</field>
    </fields>

    <!-- INSERT GROUP HEADER HERE -->

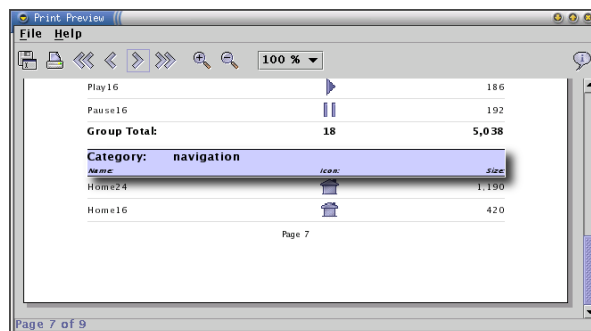
    <!-- INSERT GROUP FOOTER HERE -->

  </group>
</groups>
```

An important point to note is that the data in the **TableModel** should be *pre-sorted* according to the fields defined in the report groups, otherwise you will get some unusual results. JFreeReport does not perform any sorting itself.

As mentioned already, JFreeReport will print a *group header* every time a new group starts, and a *group footer* every time a group ends. The header and footer definitions should be inserted at the points indicated in the XML fragment above.

The group header is highlighted in the figure below:



The screenshot shows a 'Print Preview' window of a report. The report contains a table with the following data:

Name	Icon	Size
Play16		186
Pause16		192
Group Total:		5,038
Category: navigation		
Home24		1,190
Home16		420

The 'Category: navigation' row is highlighted in blue. The window also shows a toolbar with navigation icons and a zoom level of 100%. The status bar at the bottom indicates 'Page 7 of 9'.

Here is the XML used to define the group header:

```
<group-header name="anonymousBand@3921842">
  <style>
    <basic-key name="font-bold">true</basic-key>
    <basic-key name="repeat-header">false</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">30.0</basic-object>
      <basic-object name="width" class="java.lang.Double">0.0</basic-object>
    </compound-key>
    <basic-key name="font-size">12</basic-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-italic">false</basic-key>
    <basic-key name="pagebreak-before">false</basic-key>
  </style>

  <default-style>
    <basic-key name="font-bold">true</basic-key>
    <basic-key name="font-size">12</basic-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-italic">false</basic-key>
  </default-style>

  <element name="@anonymous11" type="shape/generic">
    <style>
      <basic-key name="stroke" class="java.awt.BasicStroke">0.0</basic-key>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">0.0</basic-object>
      </compound-key>
      <basic-key name="draw-shape">false</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">-100.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
      </compound-key>
      <basic-key name="keepAspectRatio">false</basic-key>
      <basic-key name="scale">true</basic-key>
      <basic-key name="paint">#ccccff</basic-key>
      <basic-key name="fill-shape">true</basic-key>
    </style>

    <datasource type="StaticDataSource">
      <compound-object name="value" class="java.awt.geom.Rectangle2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="height">100.0</basic-object>
        <basic-object name="width">100.0</basic-object>
        <basic-object name="y">0.0</basic-object>
      </compound-object>
    </datasource>
  </element>

  <element name="@anonymous12" type="shape/generic">
    <style>
      <basic-key name="stroke" class="java.awt.BasicStroke">0.5</basic-key>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">0.0</basic-object>
      </compound-key>
      <basic-key name="draw-shape">true</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">0.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
      </compound-key>
      <basic-key name="keepAspectRatio">false</basic-key>
      <basic-key name="scale">true</basic-key>
      <basic-key name="paint">black</basic-key>
      <basic-key name="fill-shape">false</basic-key>
    </style>

    <datasource type="StaticDataSource">
      <compound-object name="value" class="java.awt.geom.Line2D$Float">
        <basic-object name="x2">100.0</basic-object>
        <basic-object name="x1">0.0</basic-object>
        <basic-object name="y2">0.0</basic-object>
        <basic-object name="y1">0.0</basic-object>
      </compound-object>
    </datasource>
  </element>

  <element name="@anonymous13" type="shape/generic">
    <style>
      <basic-key name="stroke" class="java.awt.BasicStroke">0.5</basic-key>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">30.0</basic-object>
      </compound-key>
      <basic-key name="draw-shape">true</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">0.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
      </compound-key>
      <basic-key name="keepAspectRatio">false</basic-key>
      <basic-key name="scale">true</basic-key>
      <basic-key name="paint">black</basic-key>
      <basic-key name="fill-shape">false</basic-key>
    </style>

    <datasource type="StaticDataSource">
      <compound-object name="value" class="java.awt.geom.Line2D$Float">
        <basic-object name="x2">100.0</basic-object>
```

```

        <basic-object name="x1">0.0</basic-object>
        <basic-object name="y2">30.0</basic-object>
        <basic-object name="y1">30.0</basic-object>
    </compound-object>
</datasource>
</element>

<element name="@anonymous14" type="text/plain">
    <style>
        <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
            <basic-object name="x">0.0</basic-object>
            <basic-object name="y">3.0</basic-object>
        </compound-key>
        <basic-key name="dynamic_height">false</basic-key>
        <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
            <basic-object name="height" class="java.lang.Double">12.0</basic-object>
            <basic-object name="width" class="java.lang.Double">92.0</basic-object>
        </compound-key>
        <basic-key name="paint">black</basic-key>
        <basic-key name="alignment">left</basic-key>
        <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="label">
        <basic-object name="nullValue">null</basic-object>
        <basic-object name="content">Category:</basic-object>
    </template>
</element>

<element name="@anonymous15" type="text/plain">
    <style>
        <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
            <basic-object name="x">96.0</basic-object>
            <basic-object name="y">3.0</basic-object>
        </compound-key>
        <basic-key name="dynamic_height">false</basic-key>
        <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
            <basic-object name="height" class="java.lang.Double">12.0</basic-object>
            <basic-object name="width" class="java.lang.Double">120.0</basic-object>
        </compound-key>
        <basic-key name="paint">black</basic-key>
        <basic-key name="alignment">left</basic-key>
        <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="string-field">
        <basic-object name="nullValue">-</basic-object>
        <basic-object name="field">Category</basic-object>
    </template>
</element>

<element name="@anonymous16" type="text/plain">
    <style>
        <basic-key name="dynamic_height">false</basic-key>
        <basic-key name="font-size">8</basic-key>
        <basic-key name="font-bold">false</basic-key>
        <basic-key name="font-italic">true</basic-key>
        <basic-key name="paint">black</basic-key>
        <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
            <basic-object name="x">0.0</basic-object>
            <basic-object name="y">20.0</basic-object>
        </compound-key>
        <basic-key name="alignment">left</basic-key>
        <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
            <basic-object name="height" class="java.lang.Double">9.0</basic-object>
            <basic-object name="width" class="java.lang.Double">-50.0</basic-object>
        </compound-key>
        <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="label">
        <basic-object name="nullValue">null</basic-object>
        <basic-object name="content">Name:</basic-object>
    </template>
</element>

<element name="@anonymous17" type="text/plain">
    <style>
        <basic-key name="dynamic_height">false</basic-key>
        <basic-key name="font-size">8</basic-key>
        <basic-key name="font-bold">false</basic-key>
        <basic-key name="font-italic">true</basic-key>
        <basic-key name="paint">black</basic-key>
        <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
            <basic-object name="x">-50.0</basic-object>
            <basic-object name="y">20.0</basic-object>
        </compound-key>
        <basic-key name="alignment">left</basic-key>
        <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
            <basic-object name="height" class="java.lang.Double">9.0</basic-object>
            <basic-object name="width" class="java.lang.Double">-30.0</basic-object>
        </compound-key>
        <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="label">
        <basic-object name="nullValue">null</basic-object>
        <basic-object name="content">Icon:</basic-object>
    </template>
</element>

<element name="@anonymous18" type="text/plain">
    <style>

```

```

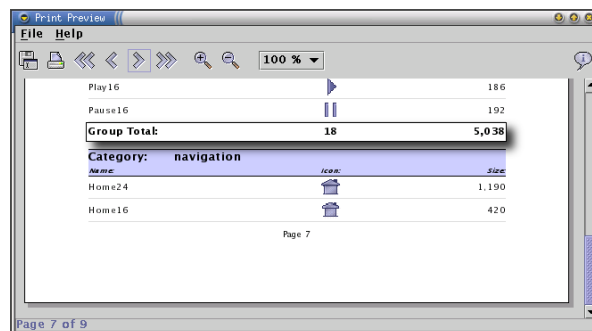
<basic-key name="dynamic_height">false</basic-key>
<basic-key name="font-size">8</basic-key>
<basic-key name="font-bold">false</basic-key>
<basic-key name="font-italic">true</basic-key>
<basic-key name="paint">black</basic-key>
<compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
  <basic-object name="x">-80.0</basic-object>
  <basic-object name="y">20.0</basic-object>
</compound-key>
<basic-key name="alignment">right</basic-key>
<compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
  <basic-object name="height" class="java.lang.Double">9.0</basic-object>
  <basic-object name="width" class="java.lang.Double">-20.0</basic-object>
</compound-key>
<basic-key name="valignment">bottom</basic-key>
</style>
<template references="label">
  <basic-object name="nullValue">null</basic-object>
  <basic-object name="content">Size</basic-object>
</template>
</element>
</group-header>

```

Some points to note:

- the **pagebreak** attribute is set to **false**. If you change this to **true**, the reporting engine will start a new page before printing the group header;
- a **string-field** template is used to display the value in the **Category** column of the report's **TableModel**;

The group footer is highlighted in the figure below:



Here is the XML used to define the group footer:

```

<group-footer name="anonymousBand08361137">
  <style>
    <basic-key name="font-bold">true</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">30.0</basic-object>
      <basic-object name="width" class="java.lang.Double">0.0</basic-object>
    </compound-key>
    <basic-key name="font-size">11</basic-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-italic">false</basic-key>
    <basic-key name="pagebreak-before">false</basic-key>
  </style>

  <default-style>
    <basic-key name="font-bold">true</basic-key>
    <basic-key name="font-size">11</basic-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-italic">false</basic-key>
  </default-style>

  <element name="@anonymous19" type="text/plain">
    <style>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">5.0</basic-object>
      </compound-key>
      <basic-key name="dynamic_height">false</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">9.0</basic-object>

```

```

        <basic-object name="width" class="java.lang.Double">100.0</basic-object>
      </compound-key>
      <basic-key name="paint">black</basic-key>
      <basic-key name="alignment">left</basic-key>
      <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="label">
      <basic-object name="nullValue">null</basic-object>
      <basic-object name="content">Group Total:</basic-object>
    </template>
  </element>

  <element name="@anonymous1a" type="text/plain">
    <style>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">-50.0</basic-object>
        <basic-object name="y">5.0</basic-object>
      </compound-key>
      <basic-key name="dynamic_height">>false</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">9.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-30.0</basic-object>
      </compound-key>
      <basic-key name="paint">black</basic-key>
      <basic-key name="alignment">center</basic-key>
      <basic-key name="valignment">bottom</basic-key>
    </style>

    <template references="number-field">
      <basic-object name="nullValue"></basic-object>
      <basic-object name="format">#0</basic-object>
      <basic-object name="field">GroupCount</basic-object>
    </template>
  </element>

  <element name="@anonymous1b" type="text/plain">
    <style>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">-80.0</basic-object>
        <basic-object name="y">5.0</basic-object>
      </compound-key>
      <basic-key name="dynamic_height">>false</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">9.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-20.0</basic-object>
      </compound-key>
      <basic-key name="paint">black</basic-key>
      <basic-key name="alignment">right</basic-key>
      <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="number-field">
      <basic-object name="nullValue"></basic-object>
      <basic-object name="format">#,##0</basic-object>
      <basic-object name="field">GroupSum</basic-object>
    </template>
  </element>
</group-footer>

```

Some points to note:

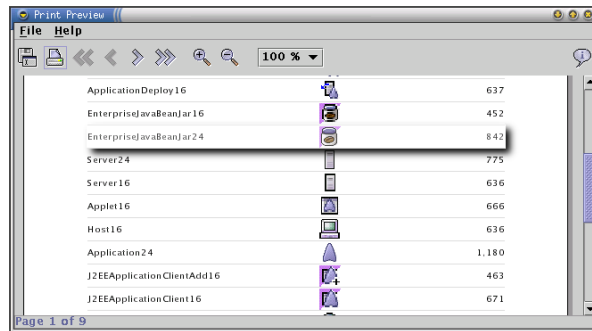
- the two elements that use **number-field** templates reference (by name) functions declared elsewhere in the report definition (see section 3.5.6);

3.5.12 The Item Band

The item band is displayed once per row of data in the report's `TableModel`. In the example, the item band contains four elements:

- a string field displaying the name of the icon;
- an image field displaying the icon itself;
- a number field displaying the size of the icon image (in bytes);
- a shape element that displays a line across the page.

One instance of the item band is highlighted in the figure below:



Here is the XML used to define the item band:

```
<!-- ===== -->
<!-- = ITEM BAND = -->
<!-- ===== -->
<itemband name="anonymousBand0581472">

  <!-- the band style -->
  <style>
    <basic-key name="font-bold">false</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">26.0</basic-object>
      <basic-object name="width" class="java.lang.Double">0.0</basic-object>
    </compound-key>
    <basic-key name="font-size">10</basic-key>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-italic">false</basic-key>
  </style>

  <!-- the default element style -->
  <default-style>
    <basic-key name="font">SansSerif</basic-key>
    <basic-key name="font-size">10</basic-key>
    <basic-key name="font-bold">false</basic-key>
    <basic-key name="font-italic">false</basic-key>
  </default-style>

  <element name="@anonymous1c" type="shape/generic">
    <style>
      <basic-key name="stroke" class="java.awt.BasicStroke">0.1</basic-key>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">25.0</basic-object>
      </compound-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">0.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-100.0</basic-object>
      </compound-key>
      <basic-key name="keepAspectRatio">false</basic-key>
      <basic-key name="scale">true</basic-key>
      <basic-key name="paint">#d9d9d9</basic-key>
      <basic-key name="draw-shape">true</basic-key>
      <basic-key name="fill-shape">false</basic-key>
    </style>
    <datasource type="StaticDataSource">
      <compound-object name="value" class="java.awt.geom.Line2D$Float">
        <basic-object name="x2">100.0</basic-object>
        <basic-object name="x1">0.0</basic-object>
        <basic-object name="y2">25.0</basic-object>
        <basic-object name="y1">25.0</basic-object>
      </compound-object>
    </datasource>
  </element>

  <!-- element to display the name from the TableModel -->
  <element name="@anonymousid" type="text/plain">
    <style>
      <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
        <basic-object name="x">0.0</basic-object>
        <basic-object name="y">8.0</basic-object>
      </compound-key>
      <basic-key name="dynamic_height">false</basic-key>
      <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
        <basic-object name="height" class="java.lang.Double">10.0</basic-object>
        <basic-object name="width" class="java.lang.Double">-50.0</basic-object>
      </compound-key>
      <basic-key name="paint">black</basic-key>
      <basic-key name="alignment">left</basic-key>
      <basic-key name="valignment">bottom</basic-key>
    </style>
    <template references="string-field">
      <basic-object name="nullValue"></basic-object>
      <basic-object name="field">Name</basic-object>
    </template>
  </element>
```

```

<!-- element to display icon from the TableModel -->
<element name="@anonymous" type="image/generic">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">-50.0</basic-object>
      <basic-object name="y">1.0</basic-object>
    </compound-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">24.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-30.0</basic-object>
    </compound-key>
    <basic-key name="keepAspectRatio">true</basic-key>
    <basic-key name="dynamic_height">false</basic-key>
    <basic-key name="scale">false</basic-key>
    <basic-key name="paint">white</basic-key>
  </style>
  <template references="image-field">
    <basic-object name="field">Icon</basic-object>
  </template>
</element>

<!-- element to display the file size from the TableModel -->
<element name="@anonymous" type="text/plain">
  <style>
    <compound-key name="absolute_pos" class="java.awt.geom.Point2D$Float">
      <basic-object name="x">-80.0</basic-object>
      <basic-object name="y">8.0</basic-object>
    </compound-key>
    <basic-key name="dynamic_height">false</basic-key>
    <compound-key name="min-size" class="com.jrefinery.report.targets.FloatDimension">
      <basic-object name="height" class="java.lang.Double">10.0</basic-object>
      <basic-object name="width" class="java.lang.Double">-20.0</basic-object>
    </compound-key>
    <basic-key name="paint">black</basic-key>
    <basic-key name="alignment">right</basic-key>
    <basic-key name="valignment">bottom</basic-key>
  </style>
  <template references="number-field">
    <basic-object name="nullValue">-</basic-object>
    <basic-object name="format">#,##0</basic-object>
    <basic-object name="field">Size</basic-object>
  </template>
</element>
</itemband>

```

Some points to note:

- unless overridden, text items in the band inherit the font settings specified in the *default-style* element for the band;
- the line element is scaled to cover the full width of the band;
- each of the remaining elements displays an item from the report's `TableModel`. The `field` attribute in the *template* element specifies the name of the column in the `TableModel` from which the data item is read.

3.6 Previewing the Report

Once the report template has been constructed, and the data is available, it is simple to create a print preview frame to display the report. The first step is to construct a `JFreeReport` instance from the XML report template file, and link the dataset to this instance. In the example, I use the following code:

```

URL in = getClass().getResource("/com/jrefinery/report/demo/swing-icons.xml");
this.report = parseReport(in);
this.report.setData(this.data);

```

This assumes that the `swing-icons.xml` file is located in the same directory as the `SwingIconsDemo.class` file.

To display the print preview frame for this report:

```

PreviewFrame frame = new PreviewFrame(this.report);
frame.setLargeIconsEnabled(true);
frame.setToolBarFloatable(false);
frame.pack();

```

```
RefineryUtilities.positionFrameRandomly(frame);  
frame.setVisible(true);  
frame.requestFocus();
```

The print preview frame then provides all the standard options to the user, including paging back and forward through the report, zooming, printing, and export to PDF.

4 Additional Documentation

4.1 Introduction

A premium version of this document, including a description of the XML report template format and reference information for the JFreeReport classes, can be purchased online (there is a link on the JFreeReport web page).

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Version 2.1, February 1999

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